

Faithful specializations of the Burau representation

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Abstract

The Burau representation, first introduced by Werner Burau in 1935, is a well-studied matrix representation of the braid group B_n with connections to the Alexander polynomial. The Burau representation of B_n is faithful for $n \leq 4$ and unfaithful for $n \geq 5$, with the longstanding case of $n = 4$ resolved by Bharathram, Birman, and Brendle in 2026. In this expository paper, we prove faithfulness for the well-known case of $n \leq 3$ before asking a more interesting question for the $n = 3$ case: is the Burau representation $\psi_3: B_3 \rightarrow GL_2(\mathbb{Z}[t, t^{-1}])$ still faithful when t is replaced with a real number? Following Scherich, we prove that it is faithful for all $t < 0$ with $t \neq -1$, while taking a surprising detour into hyperbolic geometry!

Contents

1. Introduction	2
Acknowledgments	2
2. Braid groups	3
3. The Burau representation	4
4. Faithfulness of the Burau representation	6
5. Specializations of the Burau representation	8
6. Hyperbolic geometry	10
7. Faithfulness of the specialization	13
References	15

1. Introduction

Braids are a useful algebraic tool for studying knots and links, as all links can be realized as the closure of some braid. Braids on n strands form a group B_n under composition, and we can understand the structure of B_n via its representations.

One of the best-studied representations of B_n is the Burau representation, first introduced by Werner Burau [3] in 1935. A natural question for a representation is whether it is injective (i.e. *faithful*), so that group elements can be distinguished by their actions. The Burau representation of B_n is faithful for $n \leq 4$ and unfaithful for $n \geq 5$. The $n \leq 3$ case was proven by Magnus and Peluso [5] in 1969, and the $n \geq 5$ case was proven by Bigelow [2] in 1999. Meanwhile, the case of $n = 4$ was a longstanding open problem until it was proven by Bharathram, Birman, and Brendle [1] in 2026.

In this paper, we prove faithfulness for $n \leq 3$ before asking a more interesting question for the $n = 3$ case: is the Burau representation $\psi_3: B_3 \rightarrow \mathrm{GL}_2(\mathbb{Z}[t, t^{-1}])$ still faithful when t is replaced with a real number? Such a map is called a *specialization* of the Burau representation, and Theorem 5.2 states that it is faithful for all $t < 0$ with $t \neq -1$. Following Scherich [9], we give a self-contained proof which takes a surprising detour into hyperbolic geometry!

We begin in Section 2 by introducing the braid group B_n and its connection to links and the Alexander polynomial. We define the Burau representation of B_n in Section 3 and show that it is faithful for $n \leq 3$ in Section 4. Beginning in Section 5, we discuss specializations of the Burau representation and reduce the question of faithfulness to an application of the ping-pong lemma. In Section 6, we introduce tools from hyperbolic geometry to complete the proof of faithfulness in Section 7.

Our main references are Ohtsuki [7] on the Burau representation, Scherich [9] on specializations of the Burau representation, and Mangahas [6] on hyperbolic geometry. This paper assumes familiarity with group theory, free groups, and basic knot theory, such as the definition of a knot or link.

Acknowledgments

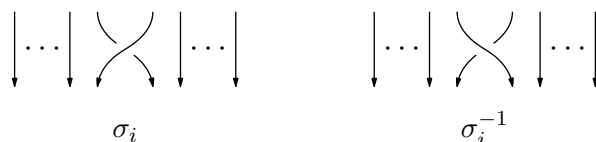
This paper was written for MIT's Seminar in Topology (18.904) in spring 2025, taught by Dr. Jonathan Zung on the topic of knot theory. I would like to thank Dr. Zung and Susan Ruff for helpful discussions, as well as Mark Saengrungkongka and Benjamin Shimabukuro for their feedback on my paper. I would also like to thank the anonymous referees from the Columbia Journal

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2. Braid groups

Intuitively, a braid is a collection of strands which weave around each other. For the purposes of this paper, we use the following definition of a braid.

Definition 2.1. Let σ_i denote a positive crossing of the i th and $(i + 1)$ th strands, and let σ_i^{-1} denote a negative crossing, as shown below.



A *braid* on n strands (up to isotopy) is obtained by vertically attaching a finite combination of σ_i 's and σ_i^{-1} 's for $1 \leq i \leq n - 1$.

Isotopy classes of braids on n strands form a group under vertical composition, called the *braid group* B_n . The braid group is known to have the following presentation.

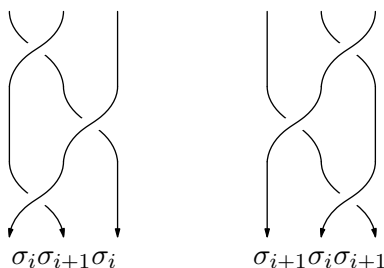
THEOREM 2.2. *The braid group B_n is presented by generators $\sigma_1, \sigma_2, \dots, \sigma_{n-1}$ with relations*

$$(2.1) \quad \sigma_i \sigma_j = \sigma_j \sigma_i \quad \text{for } |i - j| \geq 2$$

$$(2.2) \quad \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \quad \text{for } 1 \leq i \leq n - 2.$$

Products of the σ_i 's are read from left to right and correspond to composing the crossings downwards.

Intuitively, (2.1) says that crossing two pairs of strands that are far enough apart can be done in either order. Visually, (2.2) is saying that the following two braids are isotopic.



In fact B_n has the same relations as S_n , except for S_n also having $\sigma_i^2 = 1$, so B_n can be thought of as “ S_n with memory.”

There is a natural connection between braids and links, which we explain as follows.

Definition 2.3. The *closure* of a braid is the link obtained by connecting each lower end of the braid with its respective upper end, as shown in [Figure 1](#).

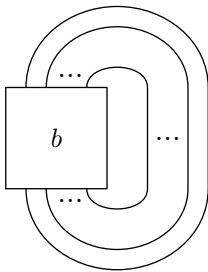


Figure 1. The closure of a braid b .

Taking the closure of any braid forms a link, and the converse is also true by Alexander’s theorem [[7](#), Theorem 2.1]: any oriented link is isotopic to the closure of some braid. However, note that this correspondence is not bijective.

3. The Burau representation

In this section, we define the Burau representation of the braid group B_n , which encodes each braid as a matrix. This is useful for computations, as matrix multiplication corresponds to composing braids. We then further the connection between braids and links in [Theorem 3.4](#), which derives the Alexander polynomial, a classical knot invariant, from the (reduced) Burau representation.

An n -dimensional *matrix representation* of a group G is a homomorphism $\rho: G \rightarrow \mathrm{GL}_n(k)$ for a field k . For our purposes, we generalize matrix representations by working with the ring $\mathbb{Z}[t, t^{-1}]$ instead of a field k , so the homomorphism will be of the form $\psi: B_n \rightarrow \mathrm{GL}_n(\mathbb{Z}[t, t^{-1}])$.

Remark 3.1. Such a homomorphism $\psi: B_n \rightarrow \mathrm{GL}_n(\mathbb{Z}[t, t^{-1}])$, when composed with an evaluation map $t \mapsto t_0$ as in [Definition 5.1](#), indeed results in a homomorphism $\rho: B_n \rightarrow \mathrm{GL}_n(\mathbb{R})$.

From the relations (2.1) and (2.2), our homomorphism $\psi: B_n \rightarrow \mathrm{GL}_n(\mathbb{Z}[t, t^{-1}])$ must satisfy

$$(3.1) \quad \psi(\sigma_i \sigma_j) = \psi(\sigma_j \sigma_i) \quad \text{for } |i - j| \geq 2$$

$$(3.2) \quad \psi(\sigma_i \sigma_{i+1} \sigma_i) = \psi(\sigma_{i+1} \sigma_i \sigma_{i+1}) \quad \text{for } 1 \leq i \leq n - 2.$$

Definition 3.2. The *unreduced Burau representation* $\tilde{\psi}_n: B_n \rightarrow \mathrm{GL}_n(\mathbb{Z}[t, t^{-1}])$ of B_n is given on generators by

$$\tilde{\psi}_n(\sigma_i) = I_{i-1} \oplus \begin{pmatrix} 1-t & t \\ 1 & 0 \end{pmatrix} \oplus I_{n-i-1}$$

for $1 \leq i \leq n - 1$, where I_k denotes the $k \times k$ identity matrix.

A direct computation shows that $\tilde{\psi}_n$ indeed satisfies (3.1) and (3.2), and is thus well-defined on B_n . The representation $\tilde{\psi}_n$ induces an action of B_n on $\mathbb{Z}[t, t^{-1}]^{\oplus n}$, and one can verify that this action is the identity on the diagonal submodule

$$D = \{(f(t), \dots, f(t)) \in \mathbb{Z}[t, t^{-1}]^{\oplus n}\}.$$

The *reduced Burau representation* ψ_n is the induced representation on the quotient $\mathbb{Z}[t, t^{-1}]^{\oplus n}/D$, and it is given explicitly by the following matrices.

Definition 3.3. The *reduced Burau representation* $\psi_n: B_n \rightarrow \mathrm{GL}_{n-1}(\mathbb{Z}[t, t^{-1}])$ of B_n for $n \geq 3$ is given on generators by

$$\psi_n(\sigma_i) = I_{i-2} \oplus \begin{pmatrix} 1 & 0 & 0 \\ t & -t & 1 \\ 0 & 0 & 1 \end{pmatrix} \oplus I_{n-i-2}$$

for $2 \leq i \leq n - 2$, and

$$\begin{aligned} \psi_n(\sigma_1) &= \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix} \oplus I_{n-3} \\ \psi_n(\sigma_{n-1}) &= I_{n-3} \oplus \begin{pmatrix} 1 & 0 \\ t & -t \end{pmatrix}. \end{aligned}$$

In the case of $n = 2$, we have $\psi_2(\sigma_1) = (-t)$.

For the remainder of this paper, we work with the reduced Burau representation ψ_n . The following theorem connects ψ_n to the Alexander polynomial, which is a polynomial associated to knots and links. It is a classical knot invariant that encodes topological information about the knot complement.

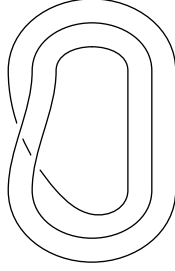
THEOREM 3.4. [7] *Let L be an oriented link, and let b be a braid on n strands with closure isotopic to L . The Alexander polynomial $\Delta_L(t)$ of L is determined by the reduced Burau representation ψ_n :*

$$\Delta_L(t) \sim \frac{1-t}{1-t^n} \det(I_{n-1} - \psi_n(b)),$$

where “ $\sim$ ” means equality up to multiplication by a unit of $\mathbb{Z}[t, t^{-1}]$.

This theorem is especially useful for efficiently computing the Alexander polynomial of a complicated braid on only a few strands. The sizes of the matrices involved depend on the number of strands rather than the number of crossings.

Example 3.5. Consider $b = \sigma_1 \sigma_2$ in B_3 , whose closure L is isotopic to the unknot, as shown below.



Then

$$\psi_n(b) = \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ t & -t \end{pmatrix} = \begin{pmatrix} 0 & -t \\ t & -t \end{pmatrix},$$

and

$$\det(I_{n-1} - \psi_n(b)) = \begin{vmatrix} 1 & t \\ -t & 1+t \end{vmatrix} = 1 + t + t^2.$$

Indeed, the Alexander polynomial for the unknot is $\frac{1-t}{1-t^3}(1+t+t^2) = 1$.

4. Faithfulness of the Burau representation

We prove the faithfulness of the Burau representation for $n \leq 3$, following Revilla Bouso [8].

THEOREM 4.1. *The Burau representation $\psi_n: B_n \rightarrow \text{GL}_{n-1}(\mathbb{Z}[t, t^{-1}])$ is faithful for $n \leq 3$.*

Proof. The $n = 1$ case is clear because B_1 is the trivial group.

For $n = 2$, we note that $B_2 \cong \mathbb{Z}$ is freely generated by σ_1 . As $\psi_2(\sigma_1) = \begin{pmatrix} -t \end{pmatrix}$, no nonzero power of $\psi_2(\sigma_1)$ equals I_1 .

In the $n = 3$ case, we have

$$(4.1) \quad \psi_3(\sigma_1) = \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix}, \quad \psi_3(\sigma_2) = \begin{pmatrix} 1 & 0 \\ t & -t \end{pmatrix}.$$

Substitute $t = -1$ to obtain the matrices $a_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $a_2 = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$. Now we compute the kernel of the composite map

$$\rho: B_3 \xrightarrow{\psi_3} \mathrm{GL}_2(\mathbb{Z}[t, t^{-1}]) \xrightarrow{t \mapsto -1} \mathrm{SL}_2(\mathbb{Z}),$$

where $\sigma_1 \mapsto a_1$ and $\sigma_2 \mapsto a_2$.

The matrices a_1, a_2 in fact generate $\mathrm{SL}_2(\mathbb{Z})$ (c.f. [8, Theorem 3.7]), and we can write the presentations of B_3 and $\mathrm{SL}_2(\mathbb{Z})$ as

$$\begin{aligned} B_3 &\cong \langle \sigma_1, \sigma_2 \mid \sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2 \rangle \\ \mathrm{SL}_2(\mathbb{Z}) &\cong \langle a_1, a_2 \mid a_1 a_2 a_1 = a_2 a_1 a_2, (a_1 a_2 a_1)^4 = 1 \rangle, \end{aligned}$$

as $a_1 a_2 a_1 = a_2 a_1 a_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. We claim that the kernel of $\rho: B_3 \rightarrow \mathrm{SL}_2(\mathbb{Z})$ is $\langle (\sigma_1 \sigma_2 \sigma_1)^4 \rangle$, and the proof follows. Indeed, from combining the facts

$$\ker \psi_3 \subset \ker \rho = \langle (\sigma_1 \sigma_2 \sigma_1)^4 \rangle$$

and $\psi_3((\sigma_1 \sigma_2 \sigma_1)^4) = \begin{pmatrix} t^6 & 0 \\ 0 & t^6 \end{pmatrix}$, we have $\ker \psi_3 = \{1\}$.

It is clear from the presentation of $\mathrm{SL}_2(\mathbb{Z})$ that $(\sigma_1 \sigma_2 \sigma_1)^4 \in \ker \rho$. For the reverse direction, suppose the word $f(\sigma_1, \sigma_2) \in B_3$ is in the kernel of ρ , so that $f(a_1, a_2) \in \mathrm{SL}_2(\mathbb{Z})$ is the identity.

Let F_{a_1, a_2} denote the free group generated by a_1, a_2 . For a subset S of a group G , let $\langle\langle S \rangle\rangle$ denote the smallest normal subgroup of G containing S . There is an isomorphism $\mathrm{SL}_2(\mathbb{Z}) \cong F_{a_1, a_2} / \langle\langle a_1 a_2 a_1 a_2^{-1} a_1^{-1} a_2^{-1}, (a_1 a_2 a_1)^4 \rangle\rangle$ following from the presentation of $\mathrm{SL}_2(\mathbb{Z})$. Then $f(a_1, a_2)$ viewed in F_{a_1, a_2} lies in $\langle\langle a_1 a_2 a_1 a_2^{-1} a_1^{-1} a_2^{-1}, (a_1 a_2 a_1)^4 \rangle\rangle$. Since the relation $\sigma_1 \sigma_2 \sigma_1 \sigma_2^{-1} \sigma_1^{-1} \sigma_2^{-1} = 1$ is already in B_3 , we can assume $f(\sigma_1, \sigma_2) \in \langle\langle (\sigma_1 \sigma_2 \sigma_1)^4 \rangle\rangle$ in B_3 . Upon verifying the relations

$$(4.2) \quad \begin{aligned} (\sigma_1 \sigma_2 \sigma_1)^2 \sigma_1 &= \sigma_1 (\sigma_1 \sigma_2 \sigma_1)^2 \\ (\sigma_1 \sigma_2 \sigma_1)^2 \sigma_2 &= \sigma_2 (\sigma_1 \sigma_2 \sigma_1)^2, \end{aligned}$$

we see that $(\sigma_1 \sigma_2 \sigma_1)^2$ is central, so $(\sigma_1 \sigma_2 \sigma_1)^4$ is as well. Then $\langle\langle (\sigma_1 \sigma_2 \sigma_1)^4 \rangle\rangle = \langle (\sigma_1 \sigma_2 \sigma_1)^4 \rangle$ and $f(\sigma_1, \sigma_2) \in \langle (\sigma_1 \sigma_2 \sigma_1)^4 \rangle$, as desired. $\square$

Remark 4.2. See Figure 2 for a visual interpretation of the computation in (4.2). Note that $\sigma_1 \sigma_2$ twists the three strands by one notch counterclockwise, so

$(\sigma_1\sigma_2)^3$ twists the three strands such that they return to their original positions. Intuitively, this twist can be done before or after any braid.



Figure 2. The element $(\sigma_1\sigma_2)^3$ in B_3 .

5. Specializations of the Burau representation

In the previous section, we showed that the Burau representation ψ_3 of B_3 is faithful. As an intermediate step, we evaluated the output of ψ_3 at $t = -1$; although the resulting representation had a nontrivial kernel, upon returning t to an indeterminate, the representation became faithful. This leads to the natural question: for which real numbers t will the resulting representation still be faithful?

Definition 5.1. A specialization ρ of the Burau representation ψ_3 is a composition

$$\rho: B_3 \xrightarrow{\psi_3} \mathrm{GL}_2(\mathbb{Z}[t, t^{-1}]) \xrightarrow{\tau} \mathrm{GL}_2(\mathbb{R}),$$

where τ is the evaluation map sending $t \mapsto t_0$ for some fixed $t_0 \in \mathbb{R}$.

The proof of [Theorem 4.1](#) implies the specialization ρ is faithful for all transcendental $t \in \mathbb{R}$. A complete classification of the faithful specializations can be found in Scherich [\[9\]](#). We prove the following illustrative case which nicely utilizes hyperbolic isometries, to be introduced in [Section 6](#).

THEOREM 5.2. *The specialization of ψ_3 is faithful for all $t < 0$ except $t = -1$.*

We use the following key lemma which gives a condition for a representation of B_n to be faithful. The proof by Long [\[4\]](#) uses theory about mapping class groups, which include braid groups B_n .

LEMMA 5.3. [4] *Let $\rho: B_n \rightarrow \text{GL}(V)$ be a representation, and suppose $N \triangleleft B_n$ is a normal subgroup which is nontrivial and noncentral. If $\rho|_N$ is faithful, then ρ is faithful, except possibly on the center of B_n .*

To apply Lemma 5.3, we use the following facts about B_3 ; the proofs are omitted, but they are standard exercises in group theory. Here, F_2 denotes the free group on two generators.

LEMMA 5.4.

- (1) *The center of B_3 is $\langle (\sigma_1\sigma_2)^3 \rangle$.*
- (2) *The subgroup $N = \langle \sigma_1^{-1}\sigma_2, \sigma_2\sigma_1^{-1} \rangle$ of B_3 is normal and isomorphic to F_2 .*

Remark 5.5. For one direction of Lemma 5.4(1), recall that in (4.2) we checked that $(\sigma_1\sigma_2)^3 = (\sigma_1\sigma_2\sigma_1)^2$ lies in the center of B_3 . Also, in Remark 4.2, we gave a visual interpretation of this fact.

Now in light of Lemma 5.3 and Lemma 5.4, we need to show that $\rho|_N$ is faithful and that ρ is faithful on the center of B_3 . The latter is easy: using (4.1), ρ sends

$$(\sigma_1\sigma_2)^3 \mapsto \begin{pmatrix} t^3 & 0 \\ 0 & t^3 \end{pmatrix},$$

and no power of $\begin{pmatrix} t^3 & 0 \\ 0 & t^3 \end{pmatrix}$ equals I_2 when $t < 0$ and $t \neq -1$ (in fact for all $t \neq 0, 1, -1$).

It remains to show that $\rho(N) = \langle \rho(\sigma_1^{-1}\sigma_2), \rho(\sigma_2\sigma_1^{-1}) \rangle$ is isomorphic to F_2 inside of $\text{GL}_2(\mathbb{R})$, so that $\rho|_N$ is faithful. We will apply the following trick for showing that a group of rank 2 is free. There is a generalization for free groups of any rank n , but we only need the case $n = 2$ in this paper.

LEMMA 5.6 (Ping-pong lemma). *Suppose a, b generate a group G which acts on a set X . Suppose there exist disjoint nonempty subsets X_a and X_b of X such that $a^k(X_b) \subset X_a$ and $b^k(X_a) \subset X_b$ for all integers $k \neq 0$. Then G is isomorphic to F_2 .*

Proof. Consider an element in G of the form $a^{i_1}b^{i_2}a^{i_3}\dots b^{i_{m-1}}a^{i_m}$ for some nonzero exponents $i_1, \dots, i_m$. Applying this element to any $x \in X_b$, we see that first $a^{i_m}(x) \in X_a$, then $b^{i_{m-1}}(a^{i_m}x) \in X_b$, and so on until, finally, $a^{i_1}(b^{i_2}\dots a^{i_m}x) \in X_a$. As X_a and X_b are disjoint, this action is nontrivial and $a^{i_1}b^{i_2}a^{i_3}\dots b^{i_{m-1}}a^{i_m}$ does not equal the identity. Furthermore, any nonempty expression in $\langle a, b \rangle$ is conjugate to an element of that form after conjugating by a sufficiently large power of a , and therefore does not equal the identity. $\square$

The name “ping-pong” comes from the alternating nature of the sets X_a and X_b . We provide a simple example of the ping-pong lemma to demonstrate its effectiveness.

Example 5.7. The subgroup generated by $a = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and $b = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ inside $\mathrm{SL}_2(\mathbb{Z})$ is isomorphic to F_2 , as shown by the ping-pong lemma. By induction, we have $a^k = \begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix}$ and $b^k = \begin{pmatrix} 1 & 0 \\ 2k & 1 \end{pmatrix}$ for all integers k . Consider the disjoint subsets of $\mathbb{R}^2$ given by

$$X_a = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid |x| > |y| \right\}, \quad X_b = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid |x| < |y| \right\}.$$

We check that $a^k(X_b) \subset X_a$ for all integers $k \neq 0$. From $\begin{pmatrix} 1 & 2k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+2ky \\ y \end{pmatrix}$, if $|x| < |y|$ then $|x+2ky| \geq |2ky| - |x| > |2ky| - |y| \geq |y|$ for all integers $k \neq 0$. Similarly, we can verify that $b^k(X_a) \subset X_b$ for all integers $k \neq 0$. Thus, $\langle a, b \rangle \cong F_2$.

However, it is challenging to find by inspection two appropriate ping-pong sets in $\mathbb{R}^2$ for the generators

$$(5.1) \quad \rho(\sigma_1^{-1}\sigma_2) = \begin{pmatrix} \frac{t-1}{t} & -1 \\ t & -t \end{pmatrix}, \quad \rho(\sigma_2\sigma_1^{-1}) = \begin{pmatrix} -\frac{1}{t} & \frac{1}{t} \\ -1 & 1-t \end{pmatrix}$$

of $\rho(N)$, especially when t is an arbitrary real number. For this problem, instead of working in $\mathbb{R}^2$, we turn to techniques from hyperbolic geometry. It turns out that the matrices from (5.1) will act on the hyperbolic plane and correspond to certain isometries, from which we can describe the ping-pong regions X_a and X_b .

6. Hyperbolic geometry

We introduce concepts from hyperbolic geometry to prove [Theorem 5.2](#). Our goal is to find a set on which the matrices from (5.1) can act.

Definition 6.1. Let $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ denote the extended complex plane. A *Möbius transformation* is a map $\widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ of the form $z \mapsto \frac{az+b}{cz+d}$ for $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$.

Remark 6.2. The condition $ad - bc \neq 0$ ensures that the map is nonconstant. One can verify that Möbius transformations form a group under composition.

It is a fact that Möbius transformations are precisely the bijective *conformal* maps $\widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ meaning that they preserve oriented angles between line segments.

More precisely, they send generalized circles to generalized circles, where a generalized circle is a circle or line.

Example 6.3. The inversion $z \mapsto \frac{1}{z}$ is a Möbius transformation. Circles passing through the origin are sent to lines, while circles not passing through the origin are sent to circles.

Next, we show that Möbius transformations can be represented by 2×2 matrices while preserving multiplication.

Definition 6.4. Let n be a positive integer and k be a field. The *projective special linear group* $\mathrm{PSL}(n, k)$ is the quotient $\mathrm{SL}(n, k)/\{I_n, -I_n\}$.

PROPOSITION 6.5. *There is an isomorphism from $\mathrm{PSL}(2, \mathbb{C})$ to the group of Möbius transformations M given by*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \left\{ z \mapsto \frac{az + b}{cz + d} \right\}.$$

Proof (sketch). After verifying that the above map is a homomorphism from $\mathrm{SL}(2, \mathbb{C})$ to M , it remains to show that the kernel is $\{I_n, -I_n\}$. Indeed if $\frac{az+b}{cz+d} = z$ for all $z \in \mathbb{C}$, then $az + b = cz^2 + dz$. Thus, $a = d$ and $b = c = 0$. From $ad - bc = 1$, we conclude that $a = d = \pm 1$. $\square$

Although this proposition is a starting point, we wish to work with matrices over $\mathbb{R}$ instead of $\mathbb{C}$, as the matrices in (5.1) are elements of $\mathrm{SL}_2(\mathbb{R})$. To obtain a similar theorem with $\mathrm{PSL}(2, \mathbb{R})$, we introduce the hyperbolic plane $\mathbb{H}^2$. We present two equivalent definitions of $\mathbb{H}^2$, first using the upper half-plane model $\mathbb{U}$, and then the Poincaré disk model $\mathbb{D}$.

Definition 6.6. Let $\mathbb{U} = \{x + iy \in \mathbb{C} \mid x, y \in \mathbb{R}, y > 0\}$ denote the *upper half-plane*.

We put a metric on $\mathbb{U}$ given by

$$(6.1) \quad (ds)^2 = \frac{(dx)^2 + (dy)^2}{y^2}$$

which allows us to compute lengths. The length of a path $f: [0, 1] \rightarrow \mathbb{U}$ where $f(t) = x(t) + iy(t)$ is thus

$$(6.2) \quad \int_0^1 \frac{1}{y(t)} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

From (6.1), the metric on $\mathbb{U}$ at each point $x + iy \in \mathbb{U}$ is the Euclidean metric $\sqrt{(dx)^2 + (dy)^2}$ scaled by $\frac{1}{y}$. Intuitively, when y is close to 0, the length of a path under this metric is much larger than with the Euclidean metric, and inversely as y increases.

Definition 6.7. A *geodesic* is a shortest path between two points. An *infinite geodesic* is a path of infinite length such that any finite subpath is a geodesic.

For two points with the same real coordinate in $\mathbb{U}$, their unique geodesic is the vertical line segment between them so that the quantity $(\frac{dx}{dt})^2$ in the length formula (6.2) is minimized. In general, the unique geodesic between two points of $\mathbb{U}$ is the arc along the circle with center on the real axis which passes through the two points [6]. This can be shown by using an appropriate Möbius transformation and the fact that Möbius transformations are conformal.

Now we are ready to state the desired theorem involving $\text{PSL}(2, \mathbb{R})$.

THEOREM 6.8 ([6]). *Every Möbius transformation that preserves $\mathbb{U}$ is an isometry of $\mathbb{U}$ with its hyperbolic metric. The group of such Möbius transformations is isomorphic to $\text{PSL}(2, \mathbb{R})$.*

The proof involves showing that such a Möbius transformation $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ must preserve the extended real line $\mathbb{R} \cup \{\infty\}$, which implies that $a, b, c, d \in \mathbb{R}$. Moreover, $\mathbb{U}$ is preserved if and only if $ad - bc > 0$ by considering where $z = i$ is sent.

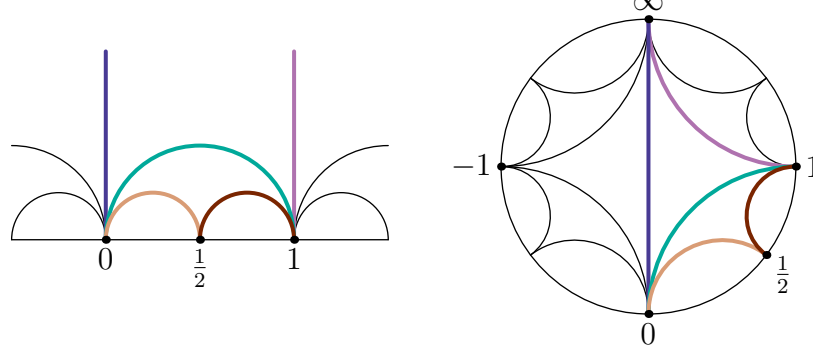
We now introduce a second model of the hyperbolic plane which will help us visualize the ping-pong lemma in our eventual proof.

Definition 6.9. The *Poincaré disk model* $\mathbb{D}$ is the image of $\mathbb{U}$ under the Möbius transformation $z \mapsto \frac{i-z}{i+z}$.

This maps the extended real line $\mathbb{R} \cup \{\infty\}$ to the unit disk in $\mathbb{C}$ and the upper half-plane into the interior. As Möbius transformations are isometries under the hyperbolic metric, the geodesics in $\mathbb{U}$ are mapped to the geodesics in $\mathbb{D}$. Consequently, the infinite geodesics in $\mathbb{D}$ are the circles and lines orthogonal to the unit circle at two points, as illustrated in Figure 3.

Finally, we need to understand how Möbius transformations act on $\mathbb{D}$. The isometries of the hyperbolic plane $\mathbb{H}^2$ induced by Möbius transformations can be classified into three types: elliptic, parabolic, and hyperbolic.

Definition 6.10. An isometry of $\mathbb{H}^2$ is *hyperbolic* if it fixes exactly two points on the boundary of $\mathbb{D}$.

Figure 3. Examples of infinite geodesics in $\mathbb{U}$ and $\mathbb{D}$.

A hyperbolic isometry has an *axis*, which is a unique geodesic in $\mathbb{D}$ on which it acts as a translation. The Möbius transformation $z \mapsto \frac{az+b}{cz+d}$ is hyperbolic if and only if $|a+d| > 2$ [6].

7. Faithfulness of the specialization

We are now ready to prove the faithfulness of the specialization in [Theorem 5.2](#) using the ping-pong lemma and tools from hyperbolic geometry.

Proof of Theorem 5.2. Define

$$a := \rho(\sigma_1^{-1}\sigma_2) = \begin{pmatrix} \frac{t-1}{t} & -1 \\ t & -t \end{pmatrix}$$

$$b := \rho(\sigma_2\sigma_1^{-1}) = \begin{pmatrix} -\frac{1}{t} & \frac{1}{t} \\ -1 & 1-t \end{pmatrix}.$$

We show that the subgroup $\langle a, b \rangle$ of $\mathrm{SL}_2(\mathbb{R})$ is isomorphic to F_2 by applying the ping-pong lemma to the action of $\langle a, b \rangle$ on $\mathbb{D}$. We see that a and b are hyperbolic isometries on $\mathbb{D}$ because both of their traces are $1 - t - \frac{1}{t} \geq 3$ by the arithmetic mean-geometric mean (AM-GM) inequality (using $t < 0$).

For the remainder of this proof, refer to [Figure 4](#) for an illustration. From the computations $a(0) = \frac{1}{t}$ and $a(1) = \infty$, the action of a on $\mathbb{D}$ sends the infinite geodesic γ_1 connecting 0 and 1 to the infinite geodesic γ_2 connecting $\frac{1}{t}$ and ∞ . Let X_a^- and X_a^+ denote the interiors of the two regions bounded by these geodesics respectively. Since a hyperbolic isometry maps connected regions to connected regions and preserves orientations, a must map $\mathbb{D} \setminus X_a^-$ (bounded by γ_1) inside X_a^+ (bounded by γ_2). Similarly, a^{-1} maps $\mathbb{D} \setminus X_a^+$ inside X_a^- .

From the computations $b(\infty) = \frac{1}{t}$ and $b(1) = 0$, the action of b sends the infinite geodesic connecting 1 and ∞ to the infinite geodesic connecting 0 and $\frac{1}{t}$. Defining X_b^- and X_b^+ analogously, b maps $\mathbb{D} \setminus X_b^-$ inside X_b^+ and b^{-1} maps $\mathbb{D} \setminus X_b^+$ inside X_b^- .

In particular, the sets $X_a = X_a^+ \cup X_a^-$ and $X_b = X_b^+ \cup X_b^-$ are disjoint and satisfy the assumptions of the ping-pong lemma by the above computations: $a^k(X_b)$ is contained in X_a^+ if $k > 0$ or X_a^- if $k < 0$. Similarly, $b^k(X_a) \subset X_b^+ \cup X_b^-$ for all $k \neq 0$. $\square$

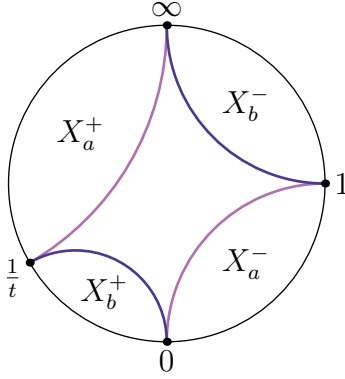


Figure 4. The ping-pong sets in the proof of [Theorem 5.2](#).

Remark 7.1. In addition to the hyperbolic isometries, another special feature of the case $t < 0$ is that $\frac{1}{t} < 0$, which produces the configuration in [Figure 4](#) consisting of four nonintersecting geodesics. Otherwise, the regions X_a and X_b would intersect.

The full classification by Scherich [\[9\]](#) which also addresses the $t \geq 0$ case compares $|\text{tr}(a)| = |\text{tr}(b)| = |1 - t - \frac{1}{t}|$ to 2 to determine whether a and b are elliptic, parabolic, or hyperbolic isometries. The specialization is faithful in the parabolic case $t = \frac{3 \pm \sqrt{5}}{2}$ and the hyperbolic case $t < \frac{3 - \sqrt{5}}{2}$ or $t > \frac{3 + \sqrt{5}}{2}$ (except at $t = 0$ and $t = -1$). The only unfaithful specializations occur in the elliptic case $\frac{3 - \sqrt{5}}{2} < t < \frac{3 + \sqrt{5}}{2}$.

The faithfulness of these specializations is a substantially stronger result than that of the Burau representation ψ_3 alone. The study of the specializations of B_3 has led to constructions of unfaithful specializations of B_4 [\[9\]](#), providing insight into the question of faithfulness for the Burau representation of B_4 .

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